Identification and Ranking of Startup Incubators in India

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1 Data Collection and Search Methodology

I began by searching for a reliable source that could provide a list of startup incubators in India. During this process, I came across the India Science, Technology and Innovation (ISTI) portal, a government website that maintains a directory of incubators across the country.

Source: India Science and Technology Portal – Incubator Listing

To collect the information systematically, I used a simple Python-based web scraping approach. The `requests` library was used to retrieve the webpage, while `pandas` was used to extract the tabular information available on the page. The extracted table was then saved as a CSV file for further analysis. The code used for this purpose is given below.

```
import requests
import pandas as pd
from io import StringIO

url = "https://www.indiascienceandtechnology.gov.in/listingpage/
   incubator"

response = requests.get(url, timeout=30)
response.raise_for_status()

tables = pd.read_html(StringIO(response.text))
df = tables[0]

# Save the extracted table as a CSV file
df.to_csv("incubator_table.csv", index=False)

print("CSV file saved successfully!")
```

This process provided an initial list of **248 incubators**, along with details such as their names, locations, establishment years, contact information, and website links.

However, these details alone were not sufficient to rank the incubators meaningfully, as they were primarily descriptive and did not provide a common numerical basis for comparison. I therefore selected four indicators that could help assess the scale of an incubator's activities and the outcomes achieved by the startups associated with it.

1. Total startups incubated
2. Number of startups graduated
3. Number of active startups
4. Total funding raised by startups (inr crores)

I then created a separate dataset containing the incubator names and used it as the basis for searching for these four indicators for 150 incubators. Although an API-based approach could potentially have made the process faster, I did not have access to a suitable free API at that time. Therefore, I carried out the searches manually and recorded the information found for each incubator.

2 Missing Data and Initial Assumptions

During the data collection process, I conducted searches across the available sources to obtain information on the four selected indicators for each startup incubator. However, despite these efforts, complete information could not be obtained for all incubators, as some incubators did not publicly disclose or provide data for all the selected indicators.

In such cases, the unavailable values were retained as missing observations rather than being replaced with zero. This decision was made because a missing value and a zero value have different interpretations. A missing value indicates that the relevant information could not be obtained from the available sources, whereas zero would imply that the incubator has no startups incubated, no startups graduated, no active startups, or no funding raised for the corresponding indicator. Assigning zero in the absence of reported information could therefore misrepresent an incubator's actual performance and introduce bias into the subsequent analysis.

Accordingly, the dataset preserves the distinction between unavailable information and an actual value of zero. This approach ensures greater transparency in data preparation and reduces the risk of misleading conclusions arising from unsupported assumptions about the performance of individual incubators. The treatment of missing observations is explicitly specified in the respective ranking methodologies to maintain the consistency and reproducibility of the analysis.

3 Methodology 1 - Borda Count-Based Ranking of Startup Incubators

3.1 Objective

The Borda Count method is used to construct an overall ranking of startup incubators based on four quantitative indicators. The method first ranks the incubators separately under each indicator and then aggregates the resulting ranks into a single score. This approach uses the relative positions of incubators rather than directly aggregating the original numerical values of the indicators.

3.2 Indicators

The following four indicators are considered:

(i) Total Startups Incubated;

(ii) Startups Graduated;

(iii) Active Startups; and

(iv) Total Funding Raised.

All four indicators are treated as benefit criteria, meaning that a higher observed value is considered preferable when constructing the ranking.

3.3 Complete-Case Selection

Let X_{ij} denote the observed value of indicator j for incubator i, where $j = 1, 2, 3, 4$.

Only incubators for which valid observations are available for all four indicators are included in the analysis. The eligible set is defined as

$$\mathcal{I} = \{i : X_{ij} \text{ is observed for all } j = 1, 2, 3, 4\}.$$

Let $N = |\mathcal{I}|$ denote the number of eligible incubators. Any incubator with a missing or non-numeric value in at least one of the four indicators is excluded from the ranking.

Missing values are not replaced with zero because the absence of a reported value does not necessarily imply that the corresponding indicator is zero. This complete-case approach ensures that every included incubator is evaluated using the same four indicators.

The resulting ranking applies only to the eligible incubators and not necessarily to the entire initial list of incubators.

3.4 Borda Count Methodology

3.4.1 Step 1: Indicator-Wise Ranking

Each eligible incubator is ranked separately for each of the four indicators. For indicator j, let r_{ij} denote the rank of incubator i.

The incubator with the highest value receives rank 1, while the incubator with the lowest value receives rank N when there are no ties.

The ranking is performed in descending order of the observed indicator values. If two or more incubators have identical values for an indicator, average ranks are assigned to the tied observations.

Thus, each incubator receives four criterion-specific ranks:

$$r_{i1}, \quad r_{i2}, \quad r_{i3}, \quad r_{i4}.$$

3.4.2 Step 2: Calculation of Borda Points

Borda points are assigned to each incubator under each indicator according to its criterion-specific rank. Since rank 1 represents the best position, the points for incubator i under indicator j are calculated as

$$B_{ij} = N - r_{ij}.$$

Consequently, an incubator with rank 1 receives $N - 1$ points, whereas an incubator with rank N receives zero points, provided there are no ties.

3.4.3 Step 3: Calculation of the Overall Borda Score

The overall Borda score is calculated by summing the points obtained across all four indicators:

$$\boxed{B_i = \sum_{j=1}^{4} B_{ij}}$$

Substituting the expression for the indicator-wise points gives

$$\boxed{B_i = \sum_{j=1}^{4} (N - r_{ij})}$$

or, equivalently,

$$\boxed{B_i = 4N - \sum_{j=1}^{4} r_{ij}.}$$

The overall score therefore aggregates the relative ranks of each incubator across the four indicators. A higher Borda score indicates a better overall position.

When there are no ties, the maximum possible score is $4(N-1)$, while the minimum possible score is zero.

3.4.4 Step 4: Final Ranking

The final ranking is obtained by arranging the eligible incubators in descending order of their overall Borda scores. The incubator with the highest score receives final rank 1.

Incubators with identical overall Borda scores receive the same competition rank. In particular, if several incubators share a score, the ranking procedure assigns them the minimum rank associated with their tied positions.

For presentation, tied incubators may be displayed in alphabetical order of their names. This secondary ordering does not change their Borda scores or their assigned ranks.

3.5 Assumptions

The methodology relies on the following assumptions:

1. **Complete observations:** Only incubators with available, valid values for all four indicators are included.

2. **Benefit criteria:** Higher values of all four indicators are treated as preferable for ranking purposes.

3. **Equal importance of indicators:** All four indicators contribute equally to the overall Borda score. No additional criterion weights are used.

4. **Ordinal aggregation:** The aggregation is based on indicator-wise ranks rather than the magnitudes of differences between indicator values.

5. **Tie handling:** Average ranks are assigned to tied indicator values. Equal overall Borda scores receive the same competition rank.

6. **Interpretation:** The ranking represents relative performance according to the four selected indicators. It does not directly measure incubator efficiency, startup survival, or overall quality.

3.6 Top 20 Incubators Based on CRITIC-MCDA

After applying the Borda Count methodology to all incubators with complete observations for the four selected indicators—Total Startups Incubated, Startups Graduated,

Active Startups, and Total Funding Raised—we obtained an overall ranking of the eligible incubators. Based on their aggregate Borda scores, the top 20 incubators are presented in Table 1.

Table 1: Top 20 Startup Incubators Based on Borda Count

Rank	Incubator Name	Borda Score
1	SIDBI Innovation & Incubation Center at Indian Institute of Technology (IIT) Kanpur	254.50
2	Nadathur S Raghavan Centre for Entrepreneurial Learning (NSRCEL) at Indian Institute of Management Bangalore (IIMB)	241.50
3	Incubation Cell at Indian Institute of Technology Madras (IIT) Madras	234.50
4	SIDBI Innovation and Incubation Center (SIIC) at Indian Institute of Technology (IIT) Kanpur	233.00
5	SIDBI Innovation and Incubation Centre (SIIC) at Indian Institute of Technology (IIT) Kanpur	229.50
6	Society for Innovation and Entrepreneurship (SINE) at Indian Institute of Technology (IIT) Bombay	224.00
7	Buisness Incubator at Indian Institute of Information Technology and Management (IIITM)	220.50
7	Indian Institute of Technology (IIT) Madras Research Park	220.50
7	Technology Business Incubator at Birla Institute of Technology and Sciences (BITS)	220.50
10	Society for Innovation and Entrepreneurship (SINE) at Indian Institute of Technology (IIT) Bombay	216.50
11	BIL-Ryerson Futures (Zone Startups)	211.00
12	Technology Business Incubator at Kalinga Institute of Industrial Technology (KIIT)	208.00
13	Global Incubation Services	202.50
14	Startup Village	198.50
15	Innovation Park (IIP) at Indian Institute of Management (IIM) Calcutta	194.00
15	Technology Business Incubator at Birla Institute of Technology and Science (BITS), Pilani	194.00
17	Rural Technology and Business Incubator (RTBI) at Indian Institute of Technology (IIT) Madras	191.00
18	STEP at Indian Institute of Technology (IIT) Kharagpur	190.00
19	Indian Angel Network (IAN) Mentoring and Incubation Services	186.00
20	Banyan Intellectual Initiatives at International Institute of Information Technology (IIIT)	183.50

3.7 Limitations

The Borda Count method is straightforward and does not require the indicators to share the same numerical scale. However, converting observations into ranks discards information about the magnitude of differences between incubators.

Furthermore, the equal treatment of the four indicators is a methodological assumption rather than evidence that all four indicators have identical substantive importance.

The complete-case restriction may also introduce selection bias if incubators with complete information differ systematically from those with missing information. Therefore, the results should be interpreted as a ranking of the eligible subset rather than a definitive ranking of all startup incubators.

4 Methodology 2 - CRITIC-Based Multi-Criteria Decision Analysis

4.1 Selection of Indicators and Data Preparation

To rank startup incubators in India, four quantitative indicators were considered: (i) Total Startups Incubated, (ii) Startups Graduated, (iii) Active Startups, and (iv) Total Funding Raised. All four indicators were treated as benefit criteria, meaning that higher values indicate better performance in the respective dimensions.

The indicator columns were converted into numeric format, with commas removed from numerical values. Empty, missing, or non-numeric entries were treated as missing observations. Only those incubators with complete observations for all four indicators were retained for further analysis. This complete-case approach ensures that all incubators are evaluated using the same set of indicators. Missing values were not replaced with zero.

Let n denote the number of eligible incubators and $m = 4$ the number of indicators. The resulting decision matrix is represented as

$$X = (x_{ij})_{n \times m},$$

where x_{ij} denotes the observed value of the j-th indicator for the i-th incubator.

4.2 Min-Max Normalization

Since the four indicators are measured on different scales, min-max normalization was applied to transform their values to the interval $[0, 1]$. For the i-th incubator and the

j-th indicator, the normalized value is defined as

$$z_{ij} = \frac{x_{ij} - \min\limits_{i} x_{ij}}{\max\limits_{i} x_{ij} - \min\limits_{i} x_{ij}}. \tag{1}$$

Here, $\min_i x_{ij}$ and $\max_i x_{ij}$ represent the minimum and maximum observed values of the j-th indicator among the eligible incubators, respectively. Since all indicators are benefit criteria, this transformation preserves the direction of preference: a higher normalized value corresponds to better performance.

Indicators with no variation across incubators cannot be normalized using this formula. Therefore, the implementation checks for constant indicators and stops the calculation if any are found.

4.3 Calculation of Standard Deviations

The standard deviation of each normalized indicator was calculated to measure its variability across incubators. For the j-th indicator, the sample standard deviation is

$$\sigma_j = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(z_{ij} - \bar{z}_j)^2}, \tag{2}$$

where

$$\bar{z}_j = \frac{1}{n}\sum_{i=1}^{n} z_{ij}.$$

A larger standard deviation indicates greater variation among incubators for that indicator, suggesting a greater capacity to distinguish between their performances.

4.4 Calculation of Inter-Criteria Conflict

The Pearson correlation coefficient between each pair of normalized indicators was calculated to assess the relationships among the criteria. Let r_{jk} denote the correlation between indicators j and k.

The conflict measure for the j-th indicator is defined as

$$C_j = \sum_{k=1}^{m}(1 - r_{jk}). \tag{3}$$

This measure captures the extent to which an indicator provides information that differs from the information supplied by the other indicators. A lower or negative correlation with another indicator contributes more to the conflict measure, whereas a strong positive correlation contributes less. Since the correlation matrix includes the correlation of each indicator with itself, the corresponding term $1 - r_{jj} = 0$ does not affect the sum.

4.5 Calculation of CRITIC Information Content

The CRITIC (Criteria Importance Through Intercriteria Correlation) method combines the variability and inter-criteria conflict of each indicator to quantify its information content. The information measure for indicator j is calculated as

$$I_j = \sigma_j C_j = \sigma_j \sum_{k=1}^{m} (1 - r_{jk}). \tag{4}$$

Thus, an indicator receives a larger information measure when it exhibits greater variability and provides information that is less redundant with the other indicators.

4.6 Determination of CRITIC Weights

The objective weight assigned to each indicator is obtained by normalizing its information measure:

$$w_j = \frac{I_j}{\sum_{k=1}^{m} I_k}, \qquad j = 1, \ldots, m. \tag{5}$$

The weights satisfy

$$w_j \geq 0, \qquad \sum_{j=1}^{m} w_j = 1. \tag{6}$$

These weights are determined from the observed data rather than assigned through subjective judgments. Consequently, the relative importance of an indicator depends on both its variability and its correlation structure with the other indicators. If the sum of the information measures is zero, the weights cannot be calculated.

4.7 Construction of the Composite CRITIC Score

After obtaining the CRITIC weights, a composite score was calculated for each incubator using the weighted sum of its normalized indicator values. The score for incubator i is

$$S_i = \sum_{j=1}^{m} w_j z_{ij}. \tag{7}$$

For the four selected indicators, this becomes

$$\begin{aligned} S_i = {} & w_1 z_{i1} + w_2 z_{i2} \\ & + w_3 z_{i3} + w_4 z_{i4}, \end{aligned} \tag{8}$$

where the weights correspond, respectively, to Total Startups Incubated, Startups Graduated, Active Startups, and Total Funding Raised.

A higher composite score indicates stronger overall performance across the four indicators, taking into account their CRITIC-derived weights.

4.8 Ranking and Selection of the Top 20 Incubators

The eligible incubators were arranged in descending order of their composite CRITIC scores. The incubator with the highest score received the first rank, followed by those with progressively lower scores. In the implementation, ties in composite scores are assigned the same minimum rank.

Finally, the 20 highest-ranked incubators were selected to produce the top-20 list. The corresponding incubator names and composite scores were reported as the final output.

4.9 Assumptions and Limitations

The methodology assumes that the four indicators are relevant measures of incubator performance and that larger values are preferable for each indicator. The weighted-sum aggregation also assumes that the normalized indicators can be combined linearly.

Although CRITIC provides objective weights based on the variability and correlation structure of the observed data, the resulting ranking depends on the selected indicators and the incubators included in the complete-case dataset. Excluding incubators with missing observations may introduce selection bias. Moreover, the four indicators primarily reflect the scale of incubation activity, startup graduation, current activity, and funding raised; they do not necessarily capture every aspect of incubator quality, such as startup survival, innovation, employment generation, or long-term impact.

4.10 Top 20 Incubators Based on CRITIC-MCDA

After applying the CRITIC-MCDA methodology to incubators with complete observations, the indicator weights were calculated, and the top 20 incubators were identified based on their composite scores. The corresponding indicator weights and top-20 ranking are presented below.

Table 2: CRITIC Weights of the Selected Indicators

Indicator	**CRITIC Weight**
Total Startups Incubated	0.2151
Startups Graduated	0.1717
Active Startups	0.4419
Total Funding Raised	0.1713
Total	**1.0000**

Table 3: Top 20 Incubators Based on the CRITIC-MCDA Method

Rank	Incubator Name	CRITIC Score
1	Nadathur S Raghavan Centre for Entrepreneurial Learning (NSRCEL) at Indian Institute of Management Bangalore (IIMB)	0.6029
2	Startup Village	0.4858
3	Incubation Cell at Indian Institute of Technology Madras (IIT) Madras	0.4216
4	SIDBI Innovation and Incubation Centre (SIIC) at Indian Institute of Technology (IIT) Kanpur	0.2634
5	Innovation Park (IIP) at Indian Institute of Management (IIM) Calcutta	0.2621
6	Indian Angel Network (IAN) Mentoring and Incubation Services	0.2499
7	SIDBI Innovation & Incubation Center at Indian Institute of Technology (IIT) Kanpur	0.2424
8	IKP Knowledge Park â€“ Life Science Incubator	0.2227
9	SIDBI Innovation and Incubation Center (SIIC) at Indian Institute of Technology (IIT) Kanpur	0.2146
10	Indian Institute of Technology (IIT) Madras Research Park	0.1724
10	Buisness Incubator at Indian Institute of Information Technology and Management (IIITM)	0.1724
12	Technology Business Incubator at Birla Institute of Technology and Sciences (BITS)	0.1676
13	BIL-Ryerson Futures (Zone Startups)	0.1510
14	Technology Business Incubator at Birla Institute of Technology and Science (BITS), Pilani	0.1509
15	Global Incubation Services	0.1390
16	Society for Innovation and Entrepreneurship (SINE) at Indian Institute of Technology (IIT) Bombay	0.1312
17	Amity Technology Incubator at Amity University	0.1308
18	Foundation for Innovation and Technology Transfer (FITT) at Indian Institute of Technology (IIT) Delhi	0.1166
19	Science & Technology Park at University of Pune	0.1115
20	International Centre for Entrepreneurship and Technology (iCreate) at Gujarat Foundation for Entrepreneurial Excellence (GFEE)	0.1097